KAPREKAR CONTEST – FINAL – SUB-JUNIOR

AMTI – OCTOBER, 2010.

Instructions:
1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Fill in FACE SLIP and your rough working should be in the answer book itself at the allotted place.
5. Maximum time allowed is THREE hours.

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1. a and b are positive integers with $a>b$. Find all pairs of a, b such that the sum of their sum, difference, product and quotient is 36.

2. The three digit number ABC, BCA, CAB are added and a four digit number with the four digits different, results. If A, B, C are distinct, find all ABC and the sum of all these three digit numbers

3. Using all the digits 1,2,3,4,5 one can form the 5-digit number 14352. It is observed that no three successive digits are in ascending order or in descending order. Find twenty such five digit numbers including the biggest and the smallest.

4. In triangle ABC, $\angle ACB=120^0$ and $\angle CAB=40^0$. AC is extended to P such that $AP=AC+2CB$. Find the measure of $\angle ABP$.

5. A faulty odometer of a car proceeds form digit 8 to the digit 0 skipping the digit 9 regardless of position of the place value. For example travelling 1km, the odometer changed for 000038 to 000040. If the odometer now reads 002010. Find the actual distance travelled by the car.

6. ABC is a triangle. BA is produced to P such the BA=AP, AC is produced to Q such that $AC=CQ$ and CB is produced to R such that $CB=BR$. The area of the triangle PQR is 2010 square units. Find the area of the triangle ABC.

7. A square piece of cloth of side 4m is cut into thin strips of width 1cm parallel to a side. These cut strips are glued together to form a long strip (ignore the length shortened by gluing the smaller strips). This long strip is made to enclose a square region along the boundary. What is the area of the region? If the long strip is made to enclose a rectangle of side 4m along the boundary find the area of the rectangular region.

8. How many two-digit natural numbers are increased by 11 when the order of the digits is reversed?
